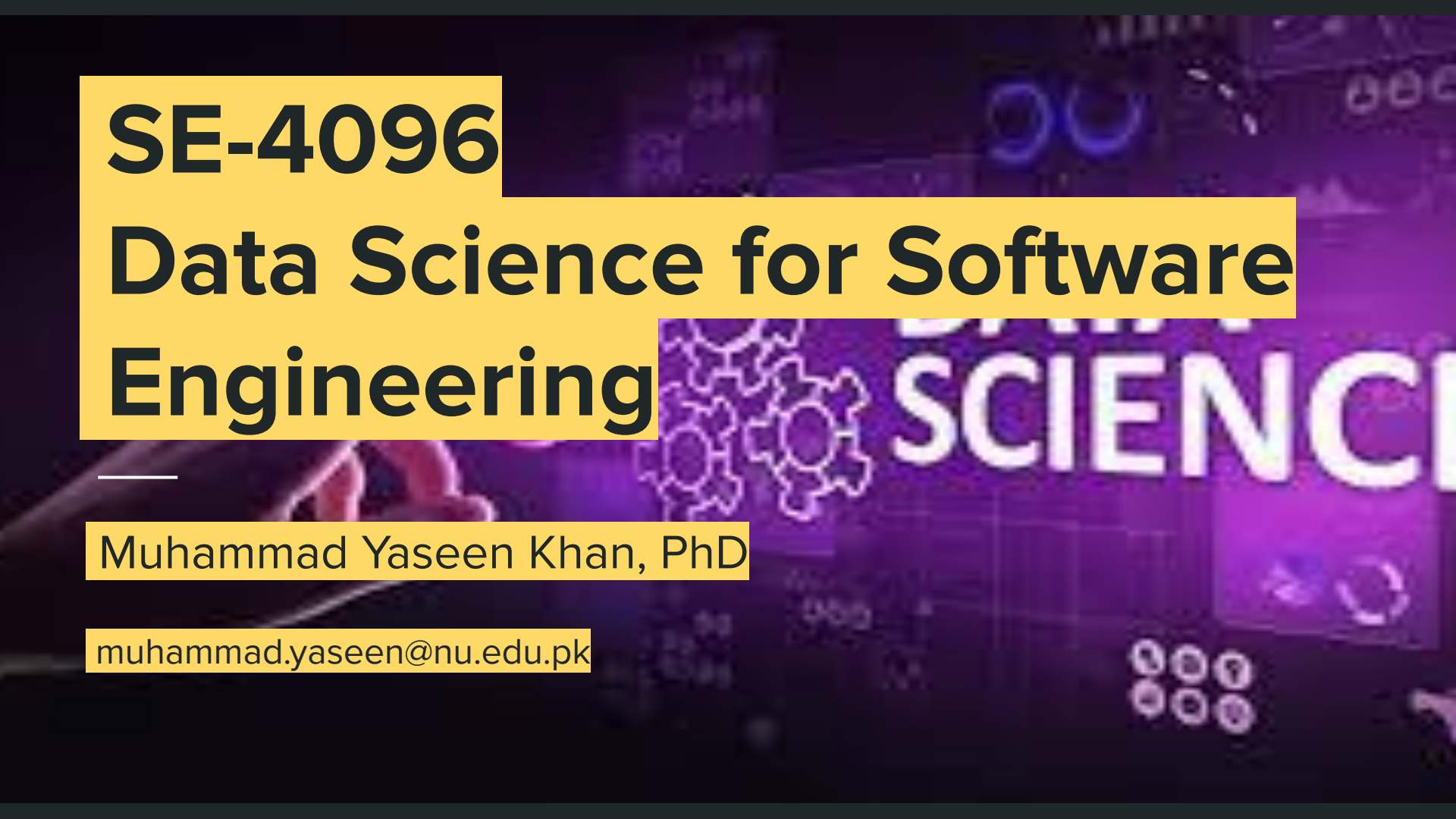




SE-4096

Data Science for Software Engineering

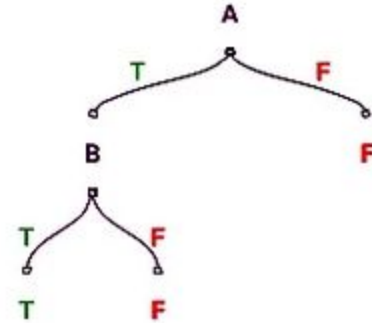
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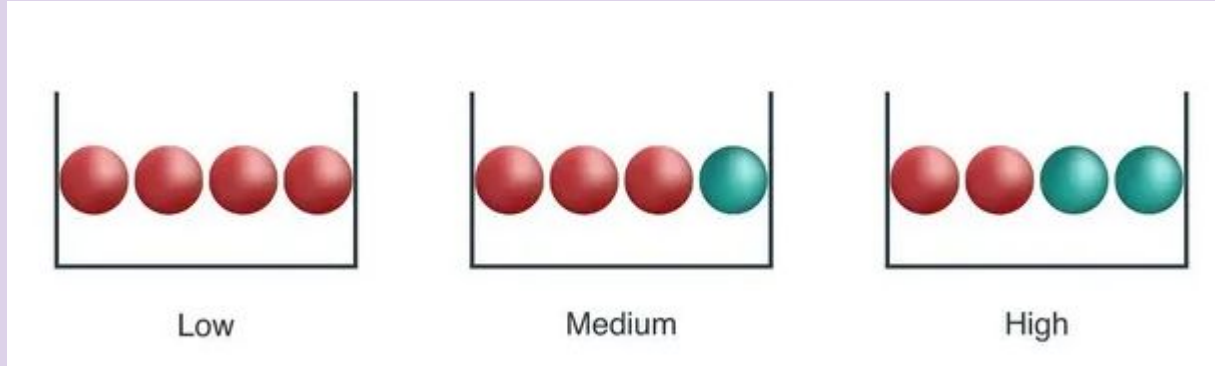
Decision Trees

How are decisions taken?

A	B	A AND B
F	F	F
F	T	F
T	F	F
T	T	T



Impurity?



From above image, a ball is randomly drawn from each bowl.

- So how much information you needed to accurately tells the color of ball?
 - left bowl needed less information as all of the ball is red colored,
 - central bowl needed more information than left bowl to tell it accurately
 - right bowl needed maximum information as both number of both color ball are same.

How can we calculate Impurity?

There are couple of impurity measures are there, we will talk about only two such measure,

1. **Entropy** is amount of information is needed to accurately describe the some sample. So if sample is homogeneous, means all the element are similar than Entropy is 0, else if sample is equally divided than entropy is maximum 1.

$$Entropy = - \sum_{i=1}^n p_i * \log(p_i)$$

2. **Gini index/ Gini impurity** is measure of inequality in sample. It has value between 0 and 1. Gini index of value 0 means sample are perfectly homogeneous and all element are similar, whereas, Gini index of value 1 means maximal inequality among elements.

$$Gini\ index = 1 - \sum_{i=1}^n p_i^2$$

How can we use Impurity?

1. So the main point is when any feature has the lowest impurity/entropy (which means high information gain) will likely to be set as top node.
2. Following this, we split the dataset as per the very feature and re-do the calculations on data splits for finding the next subsequent node.

Decision Tree Algorithms

There are Various algorithm that are used to generate decision tree from data, some are as following,

1. Classification and regression tree CART
2. ID 3
3. CHAID
4. ID 4.5

CART

Classification And Regression Trees

Developed by **Leo Breiman, Jerome Friedman, Richard Olshen, and Charles Stone.**

1. It is used for generating both classification tree and regression tree.
2. It uses Gini index as metric/cost function to evaluate split in feature selection in case of classification tree.
3. It is used for binary classification.
4. It use least square as a metric to select features in case of Regression tree.

CART

Project	Size	Scope	Experienced	Model	Buggy
P1	Big	UI	No	Agile	No
P2	Big	UI	No	Waterfall	No
P3	Moderate	UI	No	Agile	Yes
P4	Small	Web Service	Yes	Agile	Yes
P5	Small	Database	Yes	Agile	Yes
P6	Small	Database	Yes	Waterfall	No
P7	Moderate	Database	Yes	Waterfall	Yes
P8	Big	Web Service	No	Agile	No
P9	Big	Database	Yes	Agile	Yes
P10	Small	Web Service	Yes	Waterfall	Yes
P11	Big	Web Service	Yes	Waterfall	Yes
P12	Moderate	Web Service	No	Waterfall	Yes
P13	Moderate	UI	Yes	Agile	Yes
P14	Small	Web Service	No	Agile	No